

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Finalization of Electrical Power Requirement and Power Supply
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Finalization of Electrical Power Requirement and Power Supply

R01	02.06.22	Issued for Review	N.Kari	M.Ekpunobi	K.yusuff	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	02.06.22	First Issue for Clients Review
R02		
A01		

Review Code & Status (to be filled by Client)

☐ Code 1	Approved – Further work may proceed.	
☐ Code 2	Approved with comments. – Incorporate Comments. Up-Rev & Resubmit. Work May Proceed	
☐ Code 3	Reviewed – Incorporate Comments. Up-Rev & Resubmit. Work May Not Proceed	
☐ Code 4	Retained for information only	
☐ Code 5	Review Not Required – Do not Resubmit	
Date:	Position:	Sign:

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved “As-Built” documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the “As-Built” document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

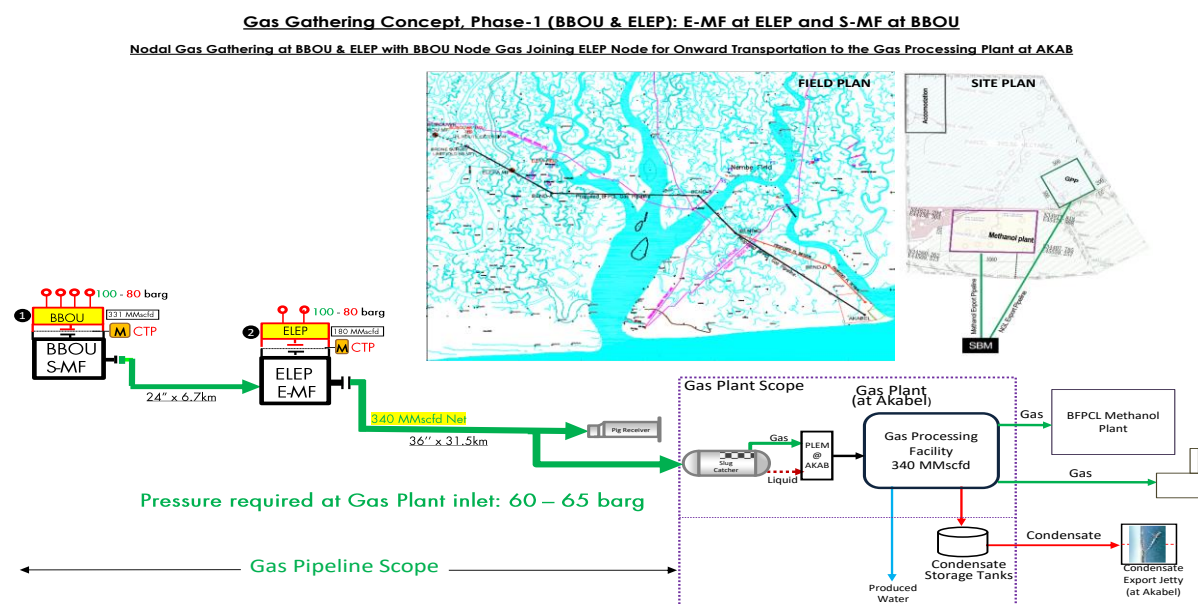


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this report is to present the outcome of the analysis conducted toward selection of power supply to the following manifold stations as part of phase 1 scope of work,

- BBOU
- ELEPA
- AKABEL

The FEED shall fully comply with all specified relevant contractual requirements.

At the conceptual stage, numerous power supply options were analyzed, schematics for these options were developed with preliminary engineering to determine material and construction cost. The analyzed options are as listed below:

1. CABLE POWER - Underground Power Cable
2. CABLE POWER - Overhead Aluminum Cable.
3. CABLE POWER - Underground Composite Cable (Data and Power)
4. SOLAR POWER - Solar Power at all Manifolds
5. CABLE & SOLAR POWER - Combination of Cable & Solar Power (Cable Power at ELEP Manifolds, Solar at BBOU, SEIB & OPUG Manifolds)
6. Cable at ELEP & NUNR - Solar at BBOU, SEIB & OPUG
7. IPP POWER - IPP at each manifold (Micro-Turbine Gas Plants, Thermo-Electric Generators (TEG))

2. FEED POWER SUPPLY OPTIONS

For the purpose of FEED, power supply to the manifold stations shall consider two most viable options listed below,

2.1. OPTION 1

Use of underground power cable from Akabel Power Plant to AKABEL Manifold station and further extension by use of RMU to ELEPA & BBOU. This option involves laying of underground cable (single or Composite) from Akabel to BBOU a total distance of circa 38.5KM with the aid of extensible RMUs and step-down to 400/230VAC at the utilization stations. This option will guarantee steady power supply and will eliminate use of generators or any other form of supply prone to vandalism.

2.2. OPTION 2

Option 2 involves use of off grid solar inverter system with back-up generators designed with auto start and stop with manual override as emergency back-up power supply to the Manifolds. One major challenge is need for a dedicated non-classified area for the installation of solar panels.

3. OPTION 1 : CABLE POWER - Underground Power Cable

Underground armored cable shall be laid from Akabel power plant to the various Manifold stations utilizing extensible RMUs & step-down transformers at the respective manifold stations. This approach will involve the following;

- 38.5km cable trench excavation.
- 38.5km laying of power underground cable (composite or single Armoured)
- Installation of step-down Transformers.
- Installation of RMUs.

Schematic of this installation is shown below:

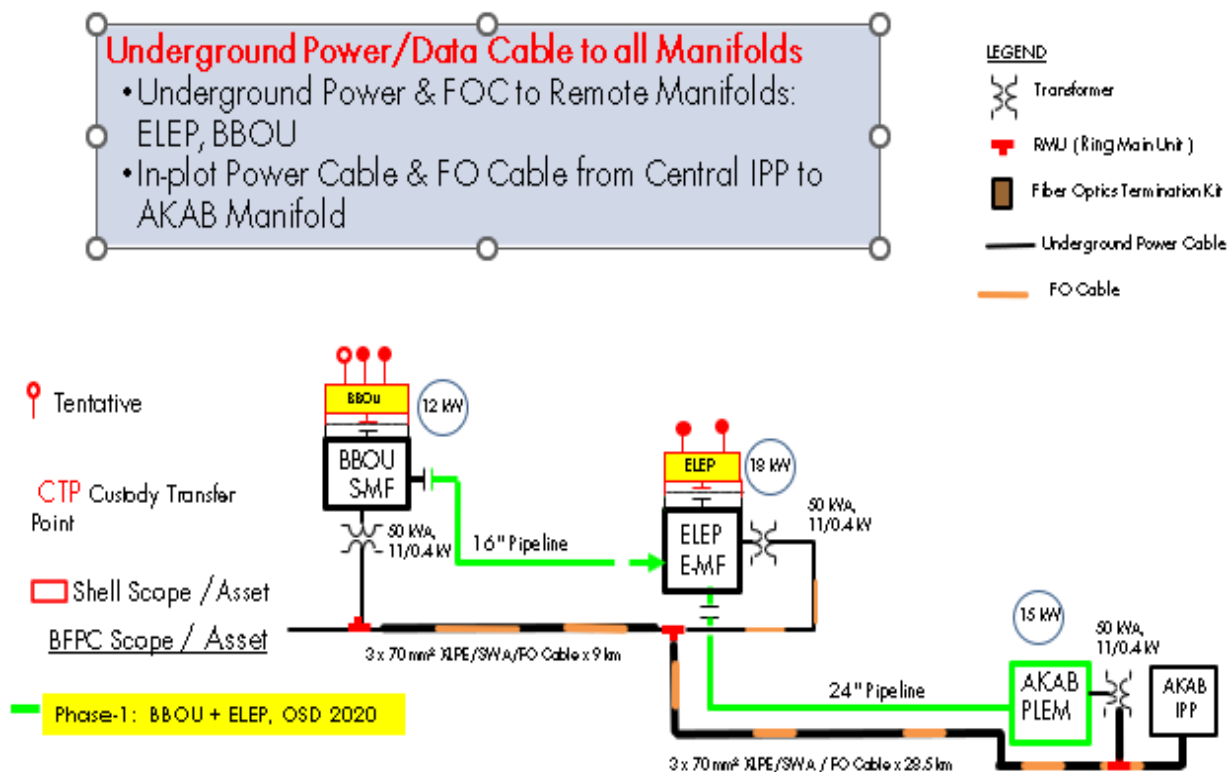


Figure 2: POWER SUPPLY OPTION 1- Underground Power Cable

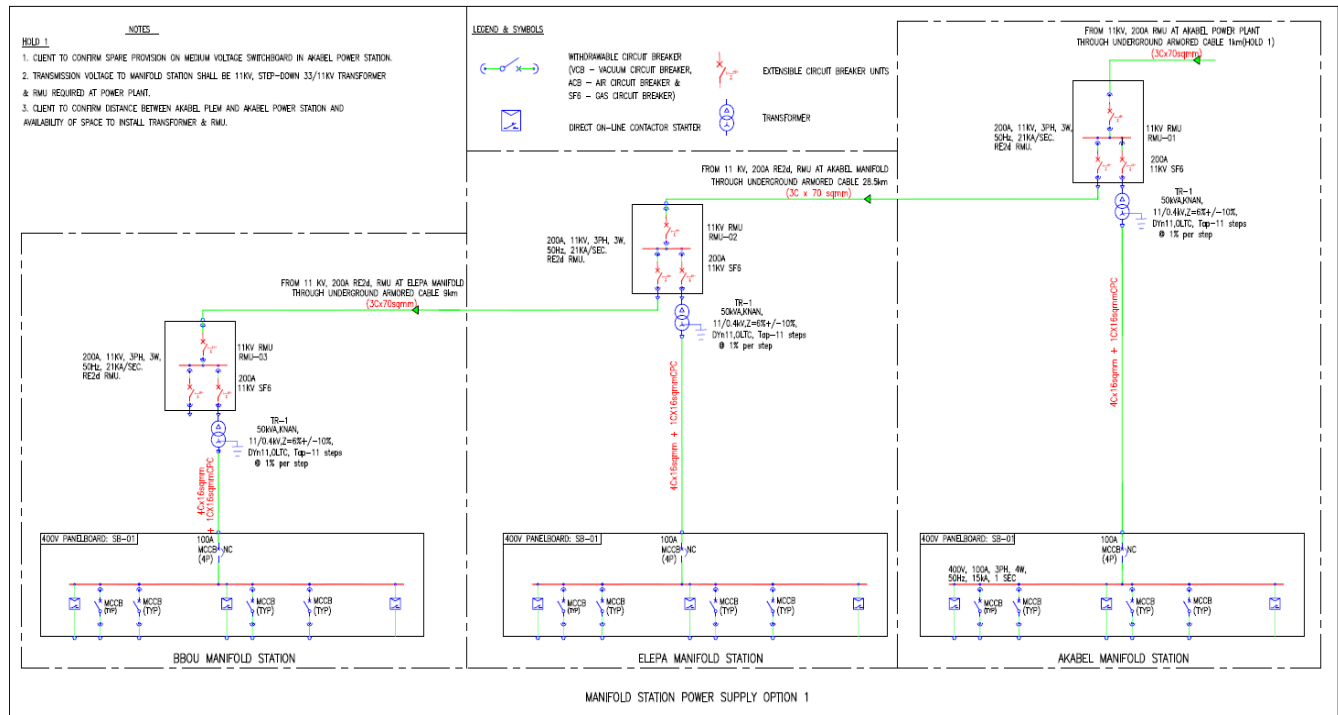


Figure3: POWER SUPPLY OPTION 1 SLD - Underground Power Cable

GAS PROCESSING AND METHANOL COMPLEX PROJECT FEED FOR GAS GATHERING PIPELINE Electrical MTO										
S/No	Material Description	Units	Quantity	Contingency	Total Quantity	Cost/Price (NGN)		Manufacturer	Model	Remarks
						Unit Rate	Total Sum			
Cables (High Voltage & Low Voltage)										
1	3C x 70mm ² Power cable	mtrs	38500	500	39000	24,500.00	955,500,000.00	NEXANS Kabelmetal or Nigerchin.	Cu/XLPE/PVC/SWA/PVC, 6.35/11kV	Armoured Underground Power Cable From AKABEL Power Plant
2	4C x 16mm ² Power cable	mtrs	150	23	175	4,000.00	698,000.00	NEXANS Kabelmetal or Nigerchin.	Cu/XLPE/PVC/SWA/PVC, 0.6/1kV	From Step-down Transformer to Distribution board.
3	70mm ² Raychem Cable jointing kit	pcs	77	12	91	12,000.00	1,086,600.00	NEXANS Kabelmetal or Nigerchin.	Cu/XLPE/PVC/SWA/PVC, 0.6/1kV	For Cable Jointing
Transformers, Substation Materials & Accessories										
4	200A, 11kV, SF6, RE2d, Ring Main Unit(RMU) with one incomers and two Outgoers.	pcs	4	0	4	4,500,000.00	18,000,000.00	ABB/Schneider Electric or approved equivalent		RMU to be installed at each Station including Akabel Power Station
5	50kVA, 11/0.4 kV, 50Hz, Dyn11, KNaN Step - down, Modular transformer complete with mounting accessories.	pcs	3	0	3	4,700,000.00	14,100,000.00	ABB/Schneider Electric or approved equivalent		
6	50kVA, 0.4/11 kV, 50Hz, YnD11, KNaN Step - up, Modular transformer complete with mounting accessories.	pcs	1	0	1	4,700,000.00	4,700,000.00	ABB/Schneider Electric or approved equivalent		
Installation, Testing & Commissioning										
7	Transformers , RMUs, installation	Lots	1	0	1		15,000,000.00			
8	Cable Trench Excavation & Cable Installation.	Lots	1	0	1		300,000,000.00			
9	Sub Total						994,084,600.00			
10	VAT (5%)									
11	Grand Total									
	NOTE 1. Ref single line diagram for connections									

Figure 4: POWER SUPPLY OPTION 1 – Cost Estimate

3.1. OPTION 1: CABLE POWER - Underground Power Cable Merits

- No impact from weather conditions, - wind, thunder, & lightning
- Neat & Tidy Installation, and not easily noticeable
- Very Reliable
- Ease of Construction
- Not highly Vulnerability to vandalization and theft

3.2. OPTION 1: CABLE POWER - Underground Power Cable Demerits

- Very Expensive
- Difficult to locate fault when it occurs.

4. OPTION 2: OFF-GRID SOLAR INVERTER & GENERATOR

An off-grid solar inverter system shall be designed to provide total power requirement on the Manifold Stations. This will be backed-up with additional Power Generator which can be powered either by gas, Methanol or diesel.

The design will include provisions for automatic start and stop with manual override to enable human intervention in case the automatic system fails.

The inverter system will be designed to guaranty enough solar panels to charge a dual redundant battery system that will support minimum of 3 days autonomy. Off-grid solar inverter system will involve the use of the following systems.

- Installation of chargers, batteries and inverters.
- Installation of PV Panels in a non-classified location.
- Standby- Generator in a non-classified location.

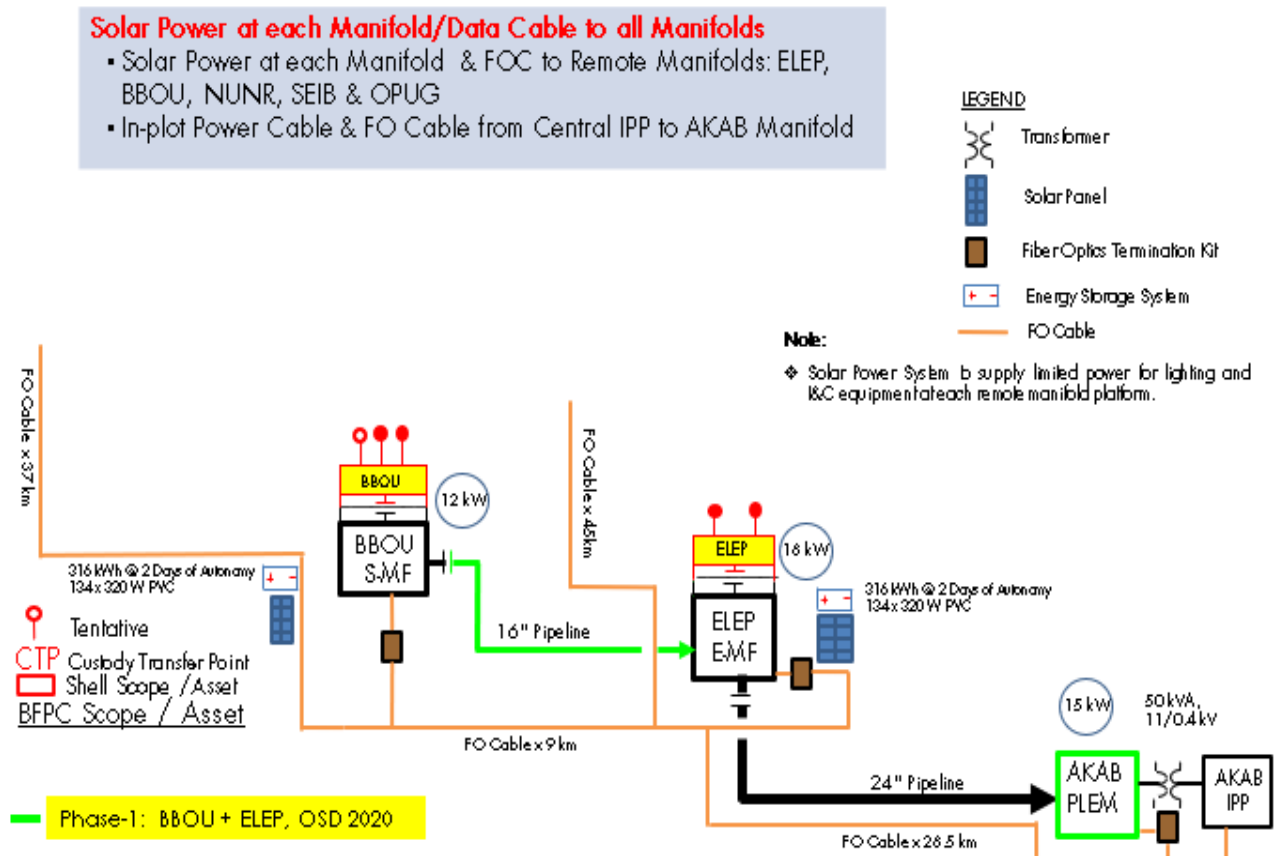


Figure 5: POWER SUPPLY OPTION 2 – Off Grid Solar Inverter

4.1. OPTION 2: OFF-GRID SOLAR INVERTER & GENERATOR

ELEPA Manifold station with the attached load list is used as case study for the design of Option 2: OFF-Grid Solar Inverter System with a total electrical load of 22kVA.

						DOCUMENT NUMBER			EA-BFPCL-GPMC-GGPL- FEED-ELE-LST-002			PROJECT:			REV	IFR	DATE:
						DOCUMENT TITLE			ELECTRICAL LOAD LIST			ELEPA MANIFOLD			PAGE	4	15 MAY 2022
S/N	TAG NO.	LOAD DESCRIPTION	SUPPLY VOLTAGE	DUTY (C/VS)	CRITICALITY (WE/N)	A	B	C	D	E	F		G		H		REMARKS
						LOAD	NOMINAL /MOTOR POWER	LOAD FACTOR A/B	EFF	POWER FACTOR	B ² C/D	(B ² C/D) X (tan θ)	B ² C/D	(B ² C/D) X (tan θ)	B ² C/D	(B ² C/D) X (tan θ)	
			[V]			[kW]	[kW]	[pu]	[pu]	[pu]	[kW]	[kVA]	[kW]	[kVA]	[kW]	[kVA]	
1	10B-01	Switchgear & Control Room Distribution Board	230	C	E	2.00	2.00	1.00	0.95	0.80	2.11	1.58	0.00	0.00	0.00	0.00	APPROXIMATED
2	10B-02	Perimeter Lighting Distribution Board	230	C	E	0.70	0.70	1.00	1.00	0.80	0.70	0.53	0.00	0.00	0.00	0.00	APPROXIMATED
3	RTU-1A	RTU PAS - ELEPA	230	C	E	1.00	1.00	1.00	0.95	0.80	1.05	0.79	0.00	0.00	0.00	0.00	APPROXIMATED
4	RTU-1B	RTU PAS (FUTURE) - ELEPA	230	C	E	1.00	1.00	1.00	0.95	0.80	1.05	0.79	0.00	0.00	0.00	0.00	APPROXIMATED
5	RTU-2A	RTU SIS - ELEPA	230	C	E	1.20	1.20	1.00	0.90	0.80	1.33	1.00	0.00	0.00	0.00	0.00	APPROXIMATED
6	RTU-2B	RTU SIS (FUTURE) - ELEPA	230	C	E	1.20	1.20	1.00	0.90	0.80	1.33	1.00	0.00	0.00	0.00	0.00	APPROXIMATED
7	HOLD	Cathodic Protection Transformer	230	C	N	5.00	5.00	1.00	0.95	0.80	5.26	3.95	0.00	0.00	0.00	0.00	APPROXIMATED
8	HOLD	Corrosion inhibitor Pump Motor A	230	C	N	1.00	1.00	1.00	0.90	0.80	1.11	0.83	0.00	0.00	0.00	0.00	APPROXIMATED
9	HOLD	Corrosion inhibitor Pump Motor B	230	S	N	1.00	1.00	1.00	0.90	0.80	0.00	0.00	0.00	0.00	1.11	0.83	APPROXIMATED
10	HOLD	SPDC Power	230	C	E	3.00	3.00	1.00	1.00	0.80	3.00	2.25	0.00	0.00	0.00	0.00	
											16.95	12.71	0.00	0.00	1.11	0.83	

SUMMARY			
	kW	kVA	kVA
Total Installed Load	18.06	13.55	22.58
Total Running Load (X%TC + Y%TI)	16.95	12.71	21.19
Peak Load (X%TC + Y%TI + Z%TS)	17.06	12.80	21.33
Where: C = Continuous Load I = Intermittent Load S = Standby Load V = V/Ital E = Emergency N = Normal TC = Total Continuous Load TI = Total Intermittent Load TS = Total Standby Load X% = 100% Y% = 30% Z% = 10%			

NOTES:

1. LOAD LIST WAS POPULATED BASED ON ASSUMED ESTIMATED LOADS	
2. LOAD ESTIMATE EXPECTED TO CHANGE CONSIDERABLE WHEN DETAILED P&ID/AJFD IS ISSUED	3. INSTRUMENTATION LOAD IS BASED ON ROUGH ESTIMATE
4. UPS SUPPLY SHALL BE VIA UPS DISTRIBUTION BOARD	5. INSTRUMENT AIR PACKAGE BASED ON ROUGH ESTIMATE

4.2. OPTION 2: OFF-GRID SOLAR INVERTER MATERIAL & COST ESTIMATE

1	QTY	DESCRIPTION	UNIT PRICE	TOTAL PRICE
			N	N
1	1	 60KVA/384VDC INVERTER	9,600,000	9,600,000.00
2	320	 230AH/12V Genus Tubular Deep Cycle Batteries	205,000	65,600,000.00
3	32	 120Amps/384v Mppt Solar Charge Controller	180,000	5,760,000.00
4	320	 Solar Panel 665W Mono Crystalline	195,000	62,400,000.00
5	80	 Aluminum Profile for locking the panel	8,500	680,000.00

6	10	 Roll of DC Flex wire of 25 mm (Red)	220,000	2,200,000.00
7	10	 Roll of DC Flex wire of 25 mm (Black)	220,000	2,200,000.00
8	10	 Roll of AC Cable of 16 mm (Red)	150,000	1,500,000.00
9	10	 Roll of AC Cable of 16 mm (Black)	150,000	1,500,000.00
10	32	 400A Single Pole DC MCB	15,000	480,000.00

11	32	 Line Surge Protector	20,000	640,000.00
12	1	 400A RMU	1,400,000	1,400,000.00
13	LOT	 LOT	LOT	1,500,000.00
14	LOT	Other Materials and Accessories (bolts, nuts, washers, connectors, bits, trunks, etc)	LOT	6,000,000.00
15	LOT	Earthing of System	LOT	600,000.00
16	LOT	Transportation and Handling	LOT	4,000,000.00
17	LOT	Labour/Commissioning	LOT	25,000,000.00
		TOTAL		191,060,000.00

4.3. OFF-GRID SOLAR INVERTER DESIGN REQUIREMENT

Real power consumption has been estimated at approximately 17.06kw which is equivalent to 21.33kva using a 0.8 power factor. Consequently, we shall be requiring a gateway energy allowance of 60kva to enable abrupt loading of 21.33kva to accommodate surge.

The surface dimensions of each of the proposed 665w solar panels proposed to be used is 2384mm x 1303mm which gives an area of 3.106m² translating to a total area of approximately

993.92m² that will be occupied by panels skewed 4 to 11 degrees to the geographical south of Nigeria with average height of 2meters from the ground in a non-classified area.

The batteries are racked in stacks of 96- batteries per rack which occupies a total area of 47m² with a maximum height of 1.5meters. This is for each of the platforms namely: ELEPA, BBOU & AKABEL.

4.4. OFF-GRID SOLAR INVERTER DESIGN REQUIREMENT

60KVA SOLAR SYSTEM will power 21.33kva load and below on abrupt loading or 40kva load on sequential loading. Duration of service per day is 24hours on the considered load of 21.33kva with a 3days back up provision for sun redundancy in cloudy weather or low sun irradiation conditions.

4.5. POWER SUPPLY OPTION 2 – Off Grid Solar Inverter Merits

- This is a renewable Energy Supply System that will be most welcome in the current campaign to reduce the use of fossil fuels and the attendant negative impact on the environment.
- Less expensive when compared with Underground Cable Power.
- Is Most suitable for unmanned remote manifold station.
- Minimum maintenance requirement.

4.6. POWER SUPPLY OPTION 2 – Off Grid Solar Inverter Demerits

- Needs a power storage system to ensure power supply during the night or periods of inadequate sunshine.
- Easily noticeable hence highly Vulnerable to vandalization and theft.
- Requires additional non classified area for installation of solar Panels as equipment is not Atex rated.

5. CONCLUSION

In view of the merit and demerits of the two options, use of underground cable power is selected as the preferred option, though initial installation cost is high but is the most reliable means of power supply to remote Manifold Stations.